

PRIMARY 2 BASIC SCIENCE SCHEME OF WORK

FIRST TERM

Week	Topic	Breakdown (Subtopics)
1	The Senses (Part 1): Sense of Sight	• Understanding our five senses • The sense of sight: using our eyes • Parts of the eye: eyelids, eyelashes, eyeball, pupil • What we can see: colors, shapes, sizes, people, animals, objects • Taking care of our eyes: washing face, not rubbing with dirty hands, enough light when reading • Things that can harm our eyes: dust, sharp objects, too much sunlight • Activity: Identifying objects by sight
2	The Senses (Part 2): Sense of Hearing and Touch	• Sense of hearing: using our ears • Sounds we hear: loud, soft, pleasant, unpleasant • Taking care of our ears: cleaning gently, avoiding loud sounds, not putting objects in ears • Sense of touch: using our skin • What we feel: hot, cold, soft, hard, rough, smooth, wet, dry • Activity: Touch box - identifying objects by touch • Matching sounds to objects
3	The Senses (Part 3): Sense of Smell and Taste	• Sense of smell: using our nose • Things we smell: nice smells (flowers, food), bad smells (rubbish, smoke) • How smell helps us: knowing if food is good or bad, enjoying flowers • Sense of taste: using our tongue

		• Different tastes: sweet, sour, bitter, salty • Parts of the tongue that taste different things • Safety: Never taste unknown substances • Activity: Identifying foods by smell and taste (with safe items)
4	Soil Types (Part 1)	• Understanding what soil is: ground we walk on, where plants grow • Where we find soil: gardens, farms, roads, forests • Uses of soil: growing plants, making pots, building houses (mud) • Types of soil: sandy soil, clay soil, loamy soil • Sandy soil: rough, does not hold water well, feels grainy • Observing and touching different soil types • Activity: Collecting soil samples from different places
5	MID-TERM EXAMINATION	Assessment of Weeks 1-4
6	MID-TERM BREAK	Holiday/Rest
7	Soil Types (Part 2)	• Clay soil: smooth, sticky when wet, holds water well, cracks when dry • Making things with clay soil: pots, bricks • Loamy soil: mixture of sand and clay, best for growing plants, dark colored • Comparing the three soil types: color, texture, water holding • Which soil is best for farming: loamy soil • Activity: Simple experiment - pouring water through different soils • Drawing and coloring different soil types
8	Air (Part 1)	• Understanding air: all around us, we cannot see it

		but we can feel it • How we know air exists: we breathe it, we feel wind, balloons inflate • Air is important: we need it to breathe and live • Things that need air: people, animals, plants • Feeling air: waving hand, fanning, blowing • Moving air is called wind • Activity: Blowing balloons, making paper fans
9	Air (Part 2)	• Properties of air: has no color (transparent), has no smell, has no taste • Air has weight: can be felt when moving • Air occupies space: fills balloons, bubbles in water • Uses of air: breathing, flying kites, drying clothes, winnowing grains • Wind helps: moving windmills, sailing boats, spreading seeds • Clean air vs. dirty air (smoke, dust) • Activity: Observing bubbles in water, flying light objects with air
10	Water (Part 1)	• Understanding water: liquid we drink, use for washing, cooking • Where we get water: rain, rivers, streams, wells, boreholes, taps • Uses of water: drinking, cooking, bathing, washing clothes/dishes, watering plants, swimming • Clean water vs. dirty water: color, smell, particles • Importance of clean water for health • Activity: Observing clean and dirty water samples (looking only)
11	REVISION	Review: The five senses, soil types, air, water

12	EXAMINATION	End of Term Assessment
13	CLOSING/VACATION	Term ends



PRIMARY 2 BASIC SCIENCE LESSON NOTES

FIRST TERM

WEEK 1: THE SENSES (PART 1): SENSE OF SIGHT.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Mention the five senses of the human body.
- Identify the eye as the organ of sight.
- List the external parts of the eye.
- State simple ways to care for the eyes.

CONTENT



Understanding Our Five Senses

Our body has special parts that help us know what is happening around us. These are called **sense organs**. Think of them as your body's five special tools. We have five senses:

- **Sight (Seeing):** using our eyes.

- **Hearing:** using our ears.
- **Smell:** using our nose.
- **Taste:** using our tongue.
- **Touch:** using our skin, especially our hands.

The Sense of Sight: Using Our Eyes

We use our eyes to see. Without eyes, everything would be dark. Our eyes are like cameras that take pictures of the world. They help us to identify:

- **Colors:** Like red apples, blue sky, and green grass.
- **Shapes:** Like circles (a plate) and squares (a window).
- **Sizes:** Big things like a car or small things like an ant.
- **People:** We can recognize our friends and family by their faces.
- **Distance:** We can see if something is near or far.

Parts of the Eye

The eye has different parts that work together like a team.

- **Eyelids:** These are the skin covers that open and close. They act like doors to protect the eyes from dust and bright light. We blink to keep our eyes wet.
- **Eyelashes:** These are the tiny hairs on the edge of the eyelids. They stop dust and sweat from falling into our eyes.
- **Eyeball:** This is the round, white part of the eye. It holds all the important parts inside.
- **Pupil:** This is the small black circle in the middle of the eye. It lets light inside so we can see. It gets bigger in the dark and smaller in bright light.

Taking Care of Our Eyes Our eyes are very delicate and important, so we must protect them.

- **Wash gently:** Wash your face and the area around your eyes with clean water every morning.
- **Avoid rubbing:** Do not rub your eyes with dirty hands, as this can bring germs inside.

- **Good lighting:** Read and do your homework with enough light. Reading in the dark makes your eyes work too hard.
- **Keep objects away:** Avoid playing with sharp objects like pencils, sticks, or toys near your eyes.
- **Never look at the sun:** Do not look directly at the sun; it is too bright and can harm the eyes. It is like looking at a very powerful torch.

EVALUATION

1. How many senses do we have?
2. What part of the body do we use for seeing?
3. Mention one thing we can see with our eyes (e.g., color, shape, person).
4. What do the eyelashes do?
5. Is it good to rub your eyes with dirty hands? (Yes/No)
6. Should we look directly at the bright sun? (Yes/No)

ASSIGNMENT

1. Draw and color a simple human eye. Label the "pupil".
 2. List two things you can see in your classroom right now.
 3. Write one important rule for taking care of your eyes.
-

WEEK 2: THE SENSES (PART 2): SENSE OF HEARING AND TOUCH.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Identify the ear as the organ for hearing.
- Differentiate between pleasant and unpleasant sounds.
- Identify the skin as the organ for touch.
- Describe how different objects feel (texture).

CONTENT



Sense of Hearing: Using Our Ears

We use our ears to hear sounds. The ears collect sounds from the environment so we can understand what is happening. Sounds are vibrations (tiny shakes) in the air that our ears catch.

Types of Sounds

- **Loud sounds:** Strong noises that you can hear from far away. Like a generator noise, shouting, or thunder.

- **Soft (quiet) sounds:** Gentle noises you hear when you are close. Like a whisper, a gentle breeze, or a cat purring.
- **Pleasant sounds:** These are nice to hear and make us happy. Like music, singing birds, or a friend laughing.
- **Unpleasant sounds (Noise):** These are disturbing and can hurt our ears. Like screaming, scratching a blackboard, or banging tables.

Taking Care of Our Ears

- **Clean gently:** Clean the outer part of your ears with a soft, damp cloth. Do not put sharp sticks, pencils, or cotton buds inside your ear. It is very dangerous.
- **Avoid loud noise:** Stay away from very loud noises (like standing near a loudspeaker) to protect your hearing.
- **Keep objects out:** Do not put beans, beads, or stones in your ears. They can get stuck.

Sense of Touch: Using Our Skin

Our skin covers our whole body. We use our skin (especially our hands and fingers) to feel things. This is the sense of touch. Our skin has nerves that send messages to our brain about what we are feeling.

What We Feel When we touch objects, we can tell:

- **Hot:** Like hot tea, a burning stove, or sunshine on a tin roof.
- **Cold:** Like ice cream, cold water, or a metal chair in the morning.
- **Soft:** Like a pillow, a teddy bear, or a cotton wool.
- **Hard:** Like a stone, a table, or a wall.
- **Rough:** Like sandpaper, a tree bark, or a grater.
- **Smooth:** Like a mirror, glass, or a polished table.

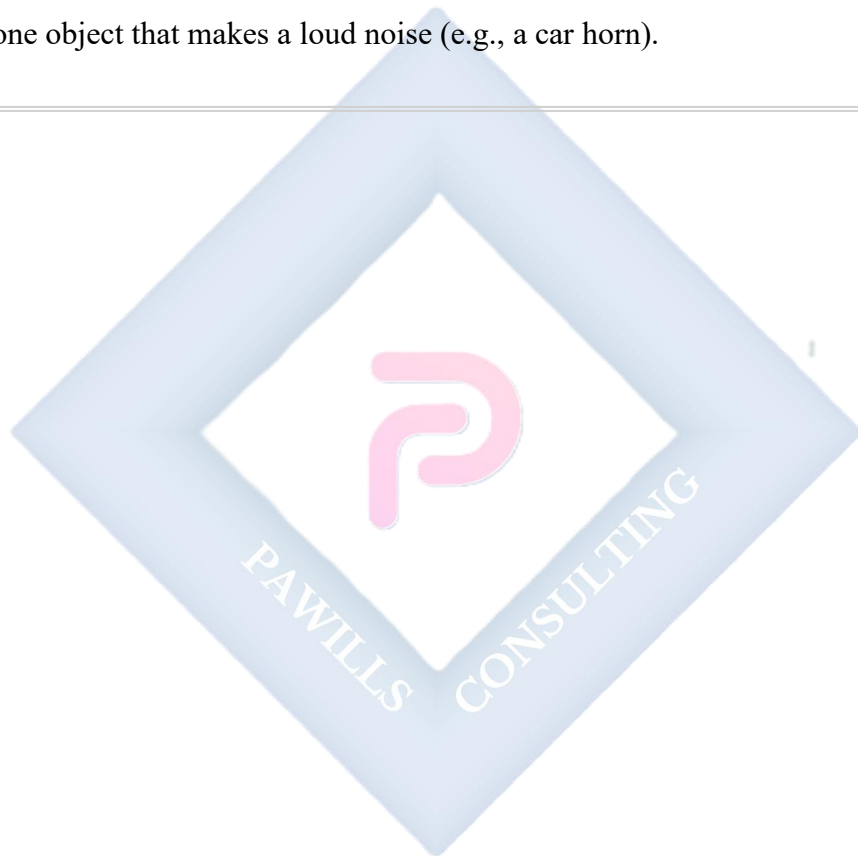
EVALUATION

1. What organ do we use for hearing?
2. Give one example of an unpleasant sound (noise).

3. Is it safe to put a sharp stick in your ear? (Yes/No)
4. What organ do we use for touching and feeling?
5. How does a stone feel? (Soft or Hard?)
6. How does ice cream feel? (Hot or Cold?)

ASSIGNMENT

1. Draw a simple picture of an ear.
 2. Touch a wall in your house. Write down if it feels smooth or rough.
 3. Name one object that makes a loud noise (e.g., a car horn).
-



WEEK 3: THE SENSES (PART 3): SENSE OF SMELL AND TASTE.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Identify the nose as the organ for smell.
- Distinguish between good and bad smells.
- Identify the tongue as the organ for taste.
- List the four basic types of tastes.

CONTENT



Sense of Smell: Using Our Nose

We use our nose to smell things in the air. The nose helps us breathe and also detects scents. Tiny particles from objects float in the air and go into our nose when we breathe in.

Types of Smells

- **Nice smells (Aromas):** These are pleasant and make us feel good.
- *Examples:* Perfume, cooked food (like jollof rice), flowers (like rose), soap.

- **Bad smells (Odors):** These are unpleasant and can warn us of danger.
- *Examples:* Rotten garbage, smoke from a fire, dirty gutters, spoilt food.
- **Importance:** Smell tells us if food is good to eat or if there is danger (like smoke from a fire). It also makes food taste better.

Sense of Taste: Using Our Tongue

We use our tongue to taste food. The tongue is inside the mouth. It has tiny bumps called **taste buds** that help us detect different tastes.

Different Tastes There are four main tastes we can feel:

- **Sweet:** Sugar, honey, cake, ice cream, ripe mango.
- **Sour:** Lemon, lime, unripe orange, tamarind.
- **Bitter:** Bitter leaf, some medicines (like chloroquine), kola nut.
- **Salty:** Salt, sea water, stock cubes.

TONGUE TASTE MAP (Simple Version)

- **Tip of tongue:** Sweet & Salty
- **Sides of tongue:** Sour
- **Back of tongue:** Bitter

Safety Rule

- Never taste or smell unknown liquids, powders, or medicines. They could be poisonous.
- Only taste food given to you by your parents or guardians.

EVALUATION

1. What part of the body is used for smelling?
2. Give one example of something that has a nice smell.
3. Give one example of something that has a bad smell.
4. What do we use to taste food?
5. What is the taste of sugar? (Sweet, Sour, Bitter, or Salty?)

6. What is the taste of bitter leaf?
7. Should you taste medicine you don't know? (Yes/No)

ASSIGNMENT

1. Draw a simple tongue in your notebook.
 2. Name one food that tastes salty.
 3. Name one item that smells like a flower (e.g., perfume, soap).
-



WEEK 4: SOIL TYPES (PART 1).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Define soil in simple terms.
- Mention three places where soil can be found.
- List two uses of soil.
- Describe the properties of sandy soil.

CONTENT



Understanding Soil

Soil is the top layer of the earth's surface. It is the soft, brown (or sometimes black or red) ground we walk on. It is made of tiny pieces of rock, dead leaves, and other materials. Plants grow their roots in soil.

Where We Find Soil Soil is found almost everywhere outdoors:

- In gardens and farms.

- Along the roadside.
- In the forest.
- Under the playground.
- In flower pots.

Uses of Soil

- **Farming:** It provides food, water, and support for plants to grow. We grow crops like maize, yam, and vegetables in soil.
- **Building:** Mud blocks (made from soil mixed with water) are used to build houses, especially in villages.
- **Pottery:** Clay soil is used to make pots, plates, and cups.

Types of Soil: Sandy Soil There are different types of soil. The first type is **Sandy Soil**.

- **Texture:** It feels rough, gritty, and grainy when you touch it. It does not stick together.
- **Water:** It does not hold water well. Water passes through it very quickly, like through a sieve.
- **Where found:** It is mostly found at the beach, in deserts, or in sandy playgrounds.
- **Growing plants:** It is not very good for growing most crops because it is too loose and dries out fast. However, some plants like carrots and watermelons can grow in sandy soil.

EVALUATION

1. What is soil? (The top layer of the earth where plants grow).
2. Mention two places where we can find soil.
3. Give one use of soil.
4. How does sandy soil feel when touched? (Rough/Smooth)
5. Does sandy soil hold water well? (Yes/No)

ASSIGNMENT

1. Go outside and touch the soil. Write one word to describe how it feels (e.g., soft, hard, rough).
2. Draw a simple plant growing in soil. Show the roots under the ground.

3. List two things humans build on or with soil (e.g., houses, mud walls).
-



WEEK 5 & 6: MID-TERM EXAMINATION & BREAK.

(Note: Lesson notes not required for examination and break weeks as per standard practice.

Proceed to Week 7.)



WEEK 7: SOIL TYPES (PART 2).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Describe the properties of clay soil.
- Describe the properties of loamy soil.
- Compare sandy, clay, and loamy soils.
- Identify the best soil for farming.

CONTENT

SOIL TYPES COMPARISON: SANDY, CLAY, & LOAMY

SANDY SOIL	CLAY SOIL	LOAMY SOIL
		
 TEXTURE: Gritty, loose, does not hold shape.	 TEXTURE: Smooth, sticky when wet, hard when dry, holds shape.	 TEXTURE: Crumbly, soft, holds moisture but drains well.
		
 COLOUR: Light brown, tan, or yellowish.	 COLOUR: Reddish-brown to grey.	 COLOUR: Dark brown to black.
		
 BEST USE: Good drainage, warming quickly; suitable for drought-tolerant plants, root vegetables.	 BEST USE: Water retention, pottery, brick-making; good for crops needing consistent moisture.	 BEST USE: Ideal for gardening and most agricultural crops; fertile and balanced.

Clay Soil

- **Texture:** It feels very smooth and powdery when dry. When wet, it becomes very sticky and can be molded like dough.
- **Water:** It holds water very well. It does not let water pass through easily, so it can become waterlogged.
- **Appearance:** It can be reddish, grey, or brown. When it dries under the sun, it cracks and becomes very hard.

- **Use:** It is good for modeling (making pots, toys) and building (mud bricks). It is not the best for most plants because it can be too heavy and sticky for roots.

Loamy Soil

- **Texture:** It is a perfect mixture of sand, clay, and decayed plants (humus). It is not too rough and not too sticky. It feels crumbly and soft.
- **Appearance:** It is usually dark brown or black in color because it is rich in nutrients (plant food).
- **Use:** This is the **best soil for planting** because it holds enough water for plants but also lets extra water drain away. It is full of food for plants.

Comparing Soil Types

TYPE	FEEL WHEN DRY	WATER HOLDING	BEST FOR
Sandy	Rough, Gritty	Poor (drains fast)	Beaches, Playgrounds
Clay	Smooth, Powdery	Very Good	Making Pots, Bricks
Loamy	Soft, Crumbly	Just Right	Farming, Gardening

EVALUATION

1. Which soil is sticky when wet?
2. Which soil is best for making pots?
3. Which soil is dark in color and best for farming?
4. Does clay soil let water pass through quickly? (Yes/No)
5. What happens to clay soil when it gets very dry? (It cracks)

ASSIGNMENT

1. Draw a simple pot made from clay soil.
2. Which soil would you choose to grow maize? (Sandy, Clay, or Loamy)
3. Why is loamy soil good for plants? (It holds water and has food)

WEEK 8: AIR (PART 1).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Define air.
- State how we know air exists.
- Define wind.
- Mention living things that need air.

CONTENT



Understanding Air

Air is a gas that is all around us. It is everywhere—inside the classroom, outside, and even in water (tiny bubbles). We cannot see it, but we can feel it.

- We **cannot see** air with our eyes because it is colorless.
- We **cannot hold** air in our hands like water.
- But we can **feel** air on our skin when it moves.

How We Know Air Exists

Even though it is invisible, we know it is there because:

- **Breathing:** We breathe air in and out every moment. Put your hand near your nose and feel the air coming out.
- **Movement:** We feel it when it moves our hair, clothes, or flags. This moving air is called wind.
- **Inflation:** When we blow air into a balloon, it gets big. The air takes up space inside the balloon.

Living Things Need Air All living things need air to stay alive. Without air, they will die.

- **People:** Need air (oxygen) to breathe.
- **Animals:** Dogs, birds, cats, and even fish (they take air from water) need air.
- **Plants:** Need air to grow and make their own food.

Wind When air is moving from one place to another, we call it **Wind**. We can create wind by waving our hands, using a fan, or blowing. Nature creates wind that blows trees, kites, and dries clothes.

EVALUATION

1. Can we see air with our eyes? (Yes/No)
2. Can we feel air? (Yes/No)
3. What is moving air called?
4. Do human beings need air to breathe? (Yes/No)
5. What happens when you blow air into a balloon? (It gets bigger)

ASSIGNMENT

1. Use a piece of paper or book to fan yourself. Describe what you feel on your skin.
 2. Draw a balloon full of air.
 3. Complete this sentence: Moving air is called _____.
-

WEEK 9: AIR (PART 2).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- State three properties of air.
- List three uses of air.
- Differentiate between clean air and dirty air.
- Mention objects that are moved by wind.

CONTENT



Properties of Air

- **Colorless:** Air has no color (it is transparent). We can see through it.
- **Odorless:** Pure air has no smell. The smells we notice come from things in the air (like perfume or smoke).
- **Tasteless:** Air has no taste.
- **Occupies Space:** Air takes up space inside things. Example: inside a tire, a football, or a balloon.
- **Has Weight:** Air has weight, though it is very light. A balloon with air is slightly heavier than an empty one.

Uses of Air and Wind

- **Breathing:** All living things breathe air to stay alive.
- **Drying:** Wind helps to dry wet clothes on the line, wet ground, and our hair.
- **Flying:** Air helps birds, airplanes, kites, and balloons to fly.
- **Cooling:** Moving air (wind or fan) cools our body when we are hot.
- **Movement:** Wind moves sailboats on the water, turns windmills to grind grain, and blows away leaves.
- **Fun:** We play with bubbles (air inside soap) and blow up balloons for parties.

Clean Air vs. Dirty Air

- **Clean Air:** Is fresh, clear, and good for our health. We find it in gardens, parks, and places with many trees.
- **Dirty Air (Polluted Air):** Contains smoke from cars/factories, dust, and bad smells. Breathing dirty air can make us cough, have allergies, or fall sick. We should avoid places with thick smoke.

EVALUATION

1. Does air have a color? (Yes/No)
2. Does air occupy space? (Yes/No) Give an example.
3. Mention two uses of air.
4. What helps wet clothes to dry? (Sun or Wind?)
5. Is smoke good for breathing? (Yes/No)
6. Mention one thing that flies in the air (e.g., bird, kite).

ASSIGNMENT

1. Draw a kite flying in the air.
 2. List two things that use air to work (e.g., football, tire).
 3. Why should we not stay near thick smoke?
-

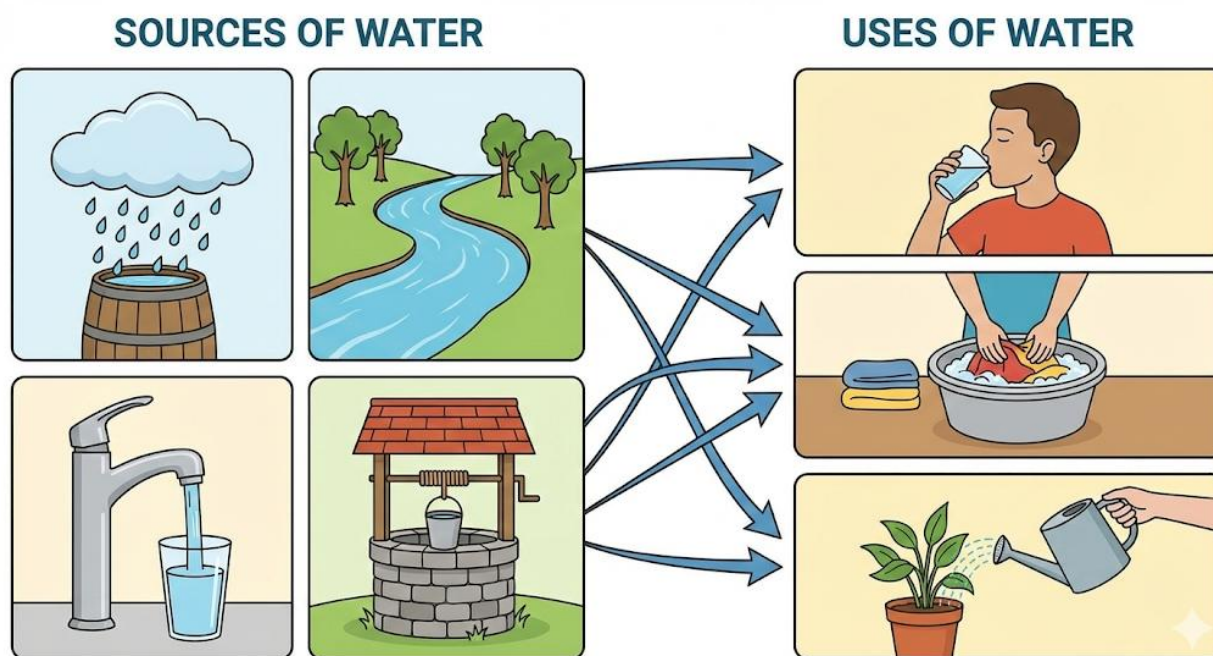
WEEK 10: WATER (PART 1).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Define water.
- List three sources of water.
- Mention four uses of water in the home.
- Differentiate between clean and dirty water.

CONTENT



Understanding Water

Water is a clear, colorless liquid that is very important for life. It has no taste or smell when pure. We drink it, and we find it in nature. No human, animal, or plant can survive without water.

Sources of Water Where do we get water from?

- **Rain:** Water falls from the sky in droplets.
- **Rivers and Streams:** Water flows on the land in channels.

- **Wells:** Deep holes dug in the ground to reach underground water.
- **Boreholes/Taps:** Pipes that bring water from underground to our houses.
- **Ocean/Sea:** Very big salty water bodies. We cannot drink this water because it is salty.
- **Spring:** Water that flows naturally from the ground.

Uses of Water We use water every day for:

- **Drinking:** to quench thirst.
- **Cooking food:** to prepare our meals.
- **Bathing and brushing teeth:** to keep our bodies clean.
- **Washing:** clothes, dishes, and cars.
- **Watering plants:** and gardens so they can grow.
- **Swimming:** and playing for fun.

Clean Water vs. Dirty Water

- **Clean Water:** Is clear (like glass), has no smell, and has no dirt particles floating in it. It is safe for washing and cooking. **For drinking, it must be boiled or treated** to kill germs.
- **Dirty Water:** Looks brown, green, or black, smells bad, and has rubbish, leaves, or worms in it. It carries germs that can make us sick with diseases like cholera and typhoid. Never drink dirty water.

EVALUATION

1. Where does rain come from? (The sky/clouds)
2. Mention two other places we find water (besides rain).
3. List three things you use water for at home.
4. What color is clean water? (Clear/Colorless)
5. Is it good to drink dirty water? (Yes/No)

ASSIGNMENT

1. Draw a bucket and a cup (tools we use to fetch water).
2. List three sources of water you see in your community.

3. Ask your parents: Is rain water clean or dirty for drinking? Write their answer.



WEEK 11: REVISION.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Recall the functions of the sense organs.
- Identify differences between soil types.
- Summarize the properties and uses of air and water.

CONTENT

Review of The Senses

- **Eyes:** for Sight (seeing colors, shapes).
- **Ears:** for Hearing (sounds loud/soft, pleasant/unpleasant).
- **Nose:** for Smell (good and bad smells).
- **Tongue:** for Taste (sweet, sour, bitter, salty).
- **Skin:** for Touch (hot, cold, soft, hard, rough, smooth).

Review of Soil

- **Sandy Soil:** Rough, gritty, drains water fast. Found at beaches.
- **Clay Soil:** Smooth, sticky when wet, holds water, cracks when dry. Good for pots.
- **Loamy Soil:** Dark, soft, crumbly. Best for farming.

Review of Air

- Air is everywhere, invisible, and has weight.
- Moving air is called **wind**.
- We need air to breathe. Uses: drying, flying, cooling.

Review of Water

- **Sources:** Rain, River, Tap, Well.
- **Uses:** Drinking, Cooking, Washing, Bathing.

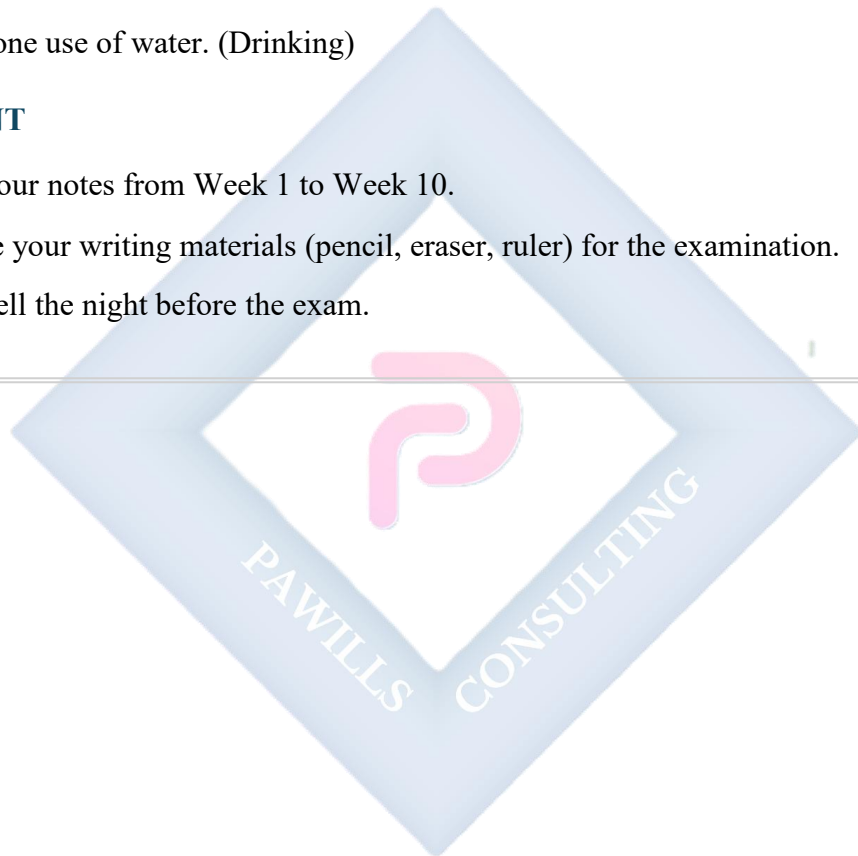
- Clean water is clear; dirty water is cloudy and unsafe.

EVALUATION

- Teacher asks mixed questions from Weeks 1--10 to prepare pupils for the end-of-term examination.
- **Example:** Which soil is best for planting? (Loamy)
- What do we use our nose for? (Smelling)
- Is air visible? (No)
- Name one use of water. (Drinking)

ASSIGNMENT

1. Read your notes from Week 1 to Week 10.
2. Prepare your writing materials (pencil, eraser, ruler) for the examination.
3. Rest well the night before the exam.



PRIMARY 2 BASIC SCIENCE SCHEME OF WORK

SECOND TERM

Week	Topic	Breakdown (Subtopics)
1	Water (Part 2)	• States of water: liquid (water we drink), solid (ice), gas (steam from boiling water) • Water becomes ice when very cold (in freezer) • Ice becomes water when warm (melting) • Water becomes steam when boiled (evaporation - simple concept) • Water cycle (very simple): rain falls, forms rivers, sun dries water, rain falls again • Activity: Observing ice melting, watching steam from hot water (teacher demonstration)
2	Colours (Part 1)	• Understanding colors: what we see with our eyes • Primary colors: red, blue, yellow (cannot be made by mixing) • Identifying primary colors in the environment: red flower, blue sky, yellow sun • Naming objects by their colors: red apple, blue car, yellow banana • Activity: Coloring with primary colors, color sorting game
3	Colours (Part 2)	• Secondary colors: green, orange, purple (made by mixing primary colors) • Mixing colors: - Red + Yellow = Orange - Blue + Yellow = Green - Red + Blue = Purple • Other colors: black, white, brown, pink, grey • Colors in nature: rainbow colors

		• Activity: Mixing paint/water colors to make new colors
4	Plants (Part 1): Parts of a Plant	• Understanding plants: living things that grow from soil • Major parts of a plant: - Roots: under the ground, hold plant in soil, take in water - Stem: above ground, holds plant upright, carries water to leaves - Leaves: green parts, make food for plant - Flowers: colorful parts, some plants have them - Fruits: some plants have fruits with seeds inside • Activity: Observing real plants, drawing and labeling plant parts
5	MID-TERM EXAMINATION	Assessment of Weeks 1-4
6	MID-TERM BREAK	Holiday/Rest
7	Plants (Part 2): Types and Uses of Plants	• Types of plants: trees (big, tall, woody stem), shrubs (smaller than trees), herbs (small, soft stem) • Examples: mango tree, hibiscus shrub, tomato plant • Uses of plants: - Food: fruits, vegetables, grains - Shelter: wood for building houses - Clothing: cotton for clothes - Medicine: herbs for treating sickness - Oxygen: plants give us fresh air to breathe • Activity: Plant walk - identifying plants in school compound
8	Plants (Part 3): How Plants Grow	• What plants need to grow: soil, water, air, sunlight

		• Stages of plant growth: seed → young plant (seedling) → adult plant • Seeds come from fruits and flowers • Caring for plants: watering, removing weeds, protecting from animals • Activity: Planting seeds in pots (beans or maize), observing growth daily
9	Animals (Part 1): Types of Animals	• Understanding animals: living things that can move from place to place • Animals around us: pets, farm animals, wild animals, insects, birds, fish • Domestic animals (pets): dog, cat, rabbit, parrot • Farm animals: goat, sheep, cow, chicken, duck, pig • Wild animals: lion, elephant, monkey, snake, crocodile • Activity: Sorting animal pictures by groups, animal sounds game
10	Animals (Part 2): Body Parts and Uses of Animals	• Body parts of animals: head, eyes, ears, legs, tail, wings (for birds) • Different animals have different body parts: fish have fins, birds have wings, snakes have no legs • Uses of animals: - Food: meat, milk, eggs - Work: donkeys carry loads, oxen plow farms - Protection: dogs guard homes - Clothing: leather from cow skin, wool from sheep • Caring for animals: feeding, giving water, shelter, treating when sick • Activity: Matching animals to their uses,

		drawing favorite animal
11	REVISION	Review: Water states, colors, plants, animals
12	EXAMINATION	End of Term Assessment
13	CLOSING/VACATION	Term ends



PRIMARY 2 BASIC SCIENCE LESSON NOTES

SECOND TERM

WEEK 1: WATER (PART 2).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Mention the three states of water (liquid, solid, gas).
- Describe how water changes to ice.
- Describe how water changes to steam.
- Explain the water cycle in simple terms.

CONTENT

States of Water

Water can be found in three different forms or states:

- **Liquid:** This is the ordinary water we drink, bathe with, and wash with. It flows easily and takes the shape of its container.
- **Solid:** When water becomes very cold (0°C or below), it freezes and turns into a hard block called **Ice**.
- **Gas:** When water becomes very hot (100°C or above), it boils and turns into an invisible gas called **Steam** or water vapour.

Changing States of Water

- **Freezing:** If we put liquid water in a freezer, it gets very cold and turns into Ice (Solid).
- *Example:* Making ice blocks.
- **Melting:** If we bring ice out of the freezer and leave it in a warm room, it turns back into water (Liquid).
- *Example:* Ice cream melting.

- **Evaporation:** If we boil water in a pot, it gets hot and turns into Steam (Gas). You can see the steam rising from the pot as a white mist.

The Water Cycle

The water cycle is how water moves around the earth in a circle. It never stops.



- **Rain falls** from the sky to the ground.
- The water flows into rivers, streams, and soaks into the soil.
- The **Sun** shines on the water and heats it up, turning it into invisible vapour (evaporation).
- The vapour goes up into the sky where it is cold and forms **clouds**.
- The clouds become heavy with water and **rain falls** again. This cycle keeps repeating.

EVALUATION

1. What are the three states of water?
2. What happens to water when we put it in a freezer? (It freezes into ____)
3. What comes out of boiling water? (Steam/Vapour)
4. When ice melts, what does it become? (Water/Liquid)
5. What dries up the water in the river? (The Sun)
6. Where does rain come from? (Clouds)

ASSIGNMENT

1. Ask your parents to show you steam when they are boiling water. (Do not touch it—it is hot!).
 2. Put a small cup of water in the freezer and check it the next day. Write down what happened.
 3. Draw rain falling from a cloud.
-



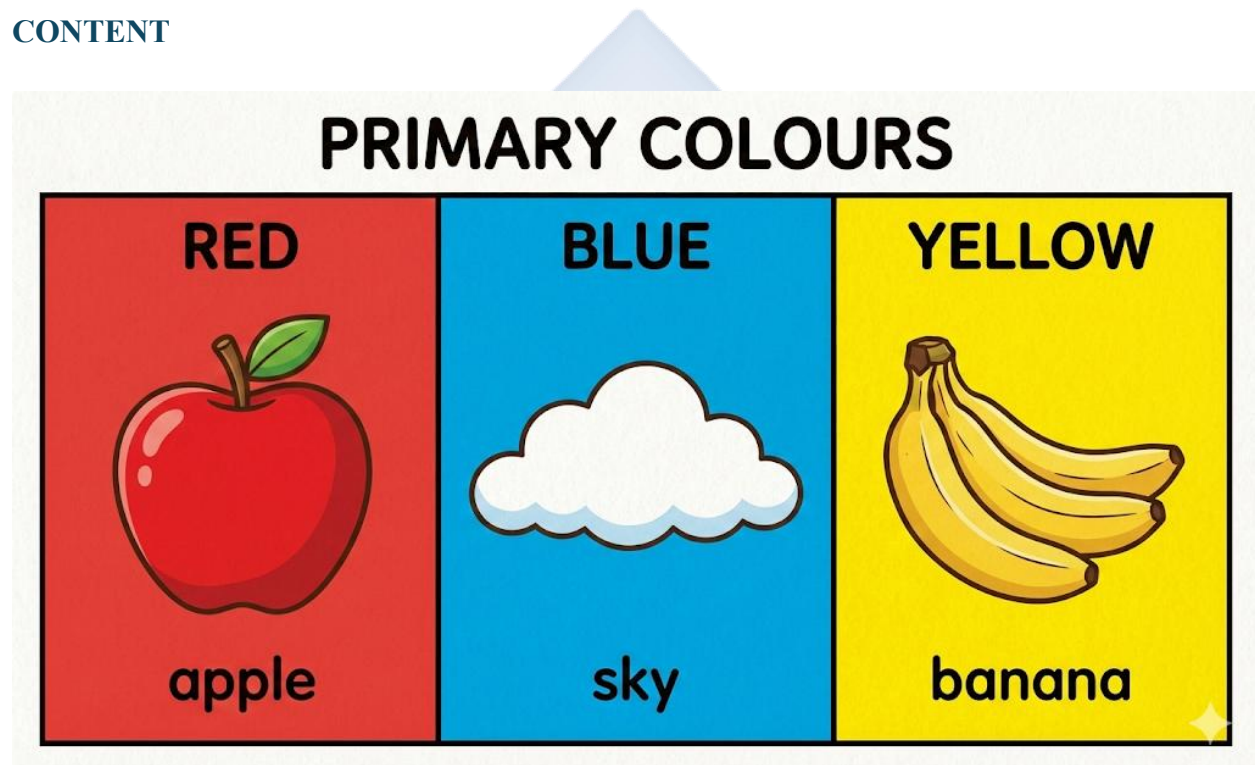
WEEK 2: COLOURS (PART 1).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Define colors as what we see with our eyes.
- List the three primary colors.
- Identify objects in the classroom or environment that match the primary colors.

CONTENT



Understanding Colours

Colours make our world look beautiful and help us tell things apart. We use our eyes to see colours. Everything around us has a colour.

Primary Colours Primary colours are the most important colours. We cannot mix other colours to get them. They are the "parents" of all other colours. There are three primary colours:

- Red
- Blue

- Yellow

Identifying Primary Colours

We can see these colours in nature and on objects:

- **Red:** A red apple, a red flower (like a rose), a red car, a red traffic light, blood.
- **Blue:** The blue sky, the blue sea, a blue pen, a blue shirt, a blue bowl.
- **Yellow:** The yellow sun, a ripe banana, a yellow yolk (inside an egg), a yellow school bus, a yellow flower.

EVALUATION

1. What part of the body do we use to see colours?
2. How many primary colours do we have?
3. List the three primary colours.
4. What is the colour of a ripe banana?
5. What is the colour of the sky on a clear day?
6. Mention one object that is red.

ASSIGNMENT

1. Draw a ball and colour it **Red**.
 2. Draw a cup and colour it **Blue**.
 3. Draw the sun and colour it **Yellow**.
-

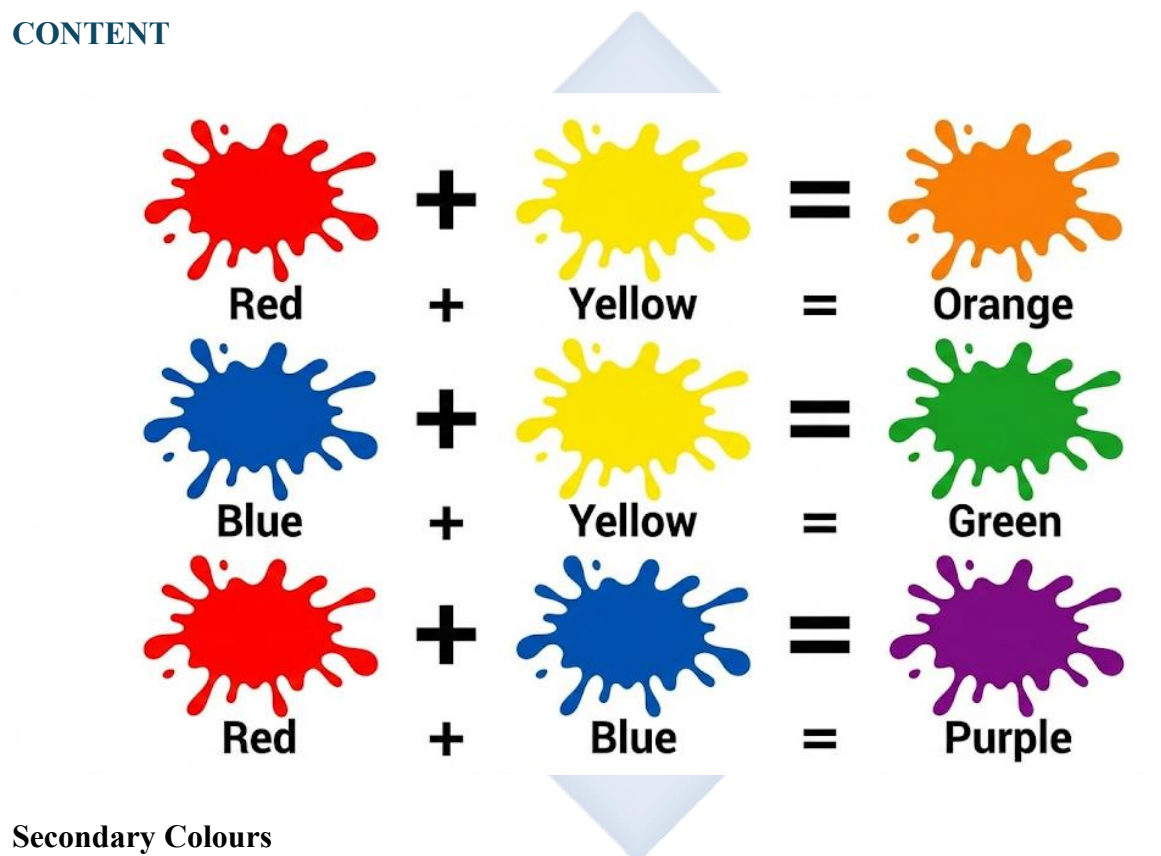
WEEK 3: COLOURS (PART 2).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Define secondary colors.
- State the result of mixing two primary colors.
- List at least three other colours found in the environment.

CONTENT



Secondary Colours

Secondary colours are the colours we get when we mix two primary colours together. They are the "children" of primary colours. The three secondary colours are:

- Green
- Orange
- Purple (sometimes called Violet)

Mixing Colours

Here is how we make secondary colours with paint:

- **Red + Yellow = Orange** (Like a carrot or an orange fruit)
- **Blue + Yellow = Green** (Like grass or leaves)
- **Red + Blue = Purple** (Like grapes or an eggplant)

Primary Mixing Chart:

- RED + YELLOW ---> ORANGE
- BLUE + YELLOW --> GREEN
- RED + BLUE -----> PURPLE

Other Colours There are many other colours we see every day. They are made by mixing primary and secondary colours in different ways.

- **White:** Like chalk, milk, paper, clouds.
- **Black:** Like charcoal, a tire, the night sky.
- **Brown:** Like the soil, wood, chocolate.
- **Pink:** Like a beautiful flower, some dresses.
- **Rainbow:** A rainbow has 7 beautiful colours in this order: Red, Orange, Yellow, Green, Blue, Indigo, Violet.

EVALUATION

1. What do we get when we mix Red and Yellow?
2. What do we get when we mix Blue and Yellow?
3. What colour is charcoal? (Black/White)
4. What colour is wood? (Brown/Green)
5. Is Green a primary or secondary colour?
6. Name one object that is Orange (e.g., orange fruit, carrot).

ASSIGNMENT

1. Find an object in your house that is Green and write its name.
2. What two colours do you mix to get Purple?

3. Draw a simple rainbow with at least three colours.



WEEK 4: PLANTS (PART 1): PARTS OF A PLANT.

OBJECTIVES

By the end of the lesson, pupils should be able to:

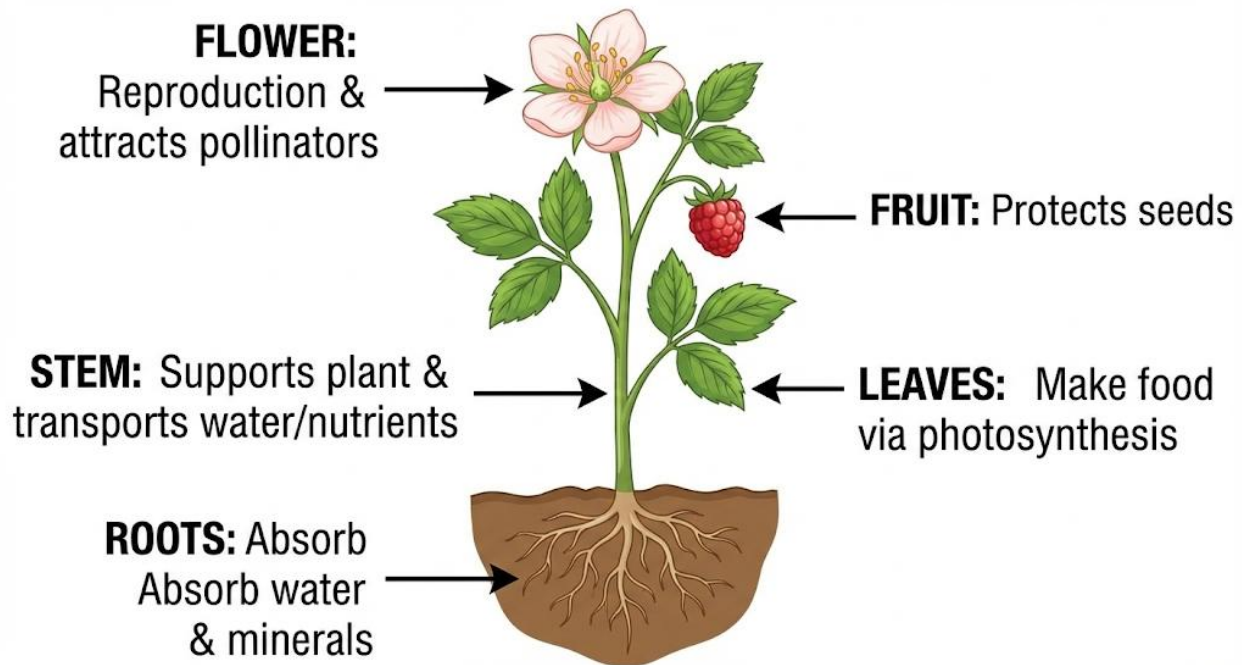
- Identify plants as living things.
- List the major parts of a plant.
- State the simple function of the roots and leaves.

CONTENT

Understanding Plants

Plants are living things. They grow from the soil. They cannot move from place to place like animals, but they breathe, feed, grow, and reproduce (make new plants).

Major Parts of a Plant A plant has different parts that help it live. The main parts are:



- **Roots:** These are usually under the ground. They are like the plant's feet and mouth.
- **Function:** They hold the plant firmly in the soil so it does not fall. They also drink water and minerals from the soil.

- **Stem:** This is the part above the ground that grows upward.
- **Function:** It holds the plant up and carries water and food between the roots and leaves. It is like a pipe or a road.
- **Leaves:** These are the green, flat parts that grow from the stem.
- **Function:** They use sunlight, air, and water to make food for the plant. This process is called **photosynthesis**.
- **Flowers:** These are the beautiful, colourful parts of the plant.
- **Function:** They attract insects and make seeds for new plants.
- **Fruits:** Some plants grow fruits after the flowers.
- **Function:** Fruits protect the seeds and help in spreading them. Fruits can be eaten by animals and people.

EVALUATION

1. Are plants living things or non-living things?
2. Which part of the plant is under the ground?
3. Which part of the plant is green and makes food?
4. What do roots take from the soil? (Water and ____)
5. Mention two parts of a plant you can see above the ground.
6. What is found inside a fruit? (Seeds)

ASSIGNMENT

1. Go outside and gently touch the leaf of a plant. Is it hard or soft? Write your answer.
 2. Draw a simple plant and label the **Root** and **Stem**.
 3. Name one fruit you like to eat.
-

WEEK 5 & 6: MID-TERM EXAMINATION AND MID-TERM BREAK.

(Note: No lesson notes are required for these weeks as they are for assessment and holiday.)



WEEK 7: PLANTS (PART 2): TYPES AND USES OF PLANTS.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Mention the three types of plants (Trees, Shrubs, Herbs).
- Give one example for each type of plant.
- List four uses of plants to human beings.

CONTENT

PLANTS: TYPES & USES (FOOD, WOOD, MEDICINE)		
TREES (Large, woody, single trunk)	SHRUBS (Medium, woody, multiple stems)	HERBS (Small, non-woody, soft stems)
 Examples: Oak, Mango, Neem, Pine	 Examples: Rose, Hibiscus, Tulsi, Lemon	 Examples: Mint, Coriander, Spinach, Basil
USES:  WOOD (Timber, Construction, Furniture)  FOOD (Fruits, Nuts)  MEDICINE (Bark, Leaves, Oil)	 FOOD (Fruits, Flavoring)  WOOD (Fuel, Crafts)  MEDICINE (Leaves, Flowers, Roots)	 FOOD (Leafy Greens, Spices, Flavoring)  WOOD (Not applicable)  MEDICINE (Leaves, Extracts, Oils)
Plants provide essential resources for human survival and well-being through their diverse forms and properties.		

Types of Plants

Plants come in different sizes and shapes. We can group them into three main types:

- **Trees:** These are very big and tall plants. They have a strong, hard stem called a **trunk** and many branches.
- *Examples:* Mango tree, Coconut tree, Iroko tree, Palm tree.
- They live for many years.

- **Shrubs:** These are medium-sized plants. They are smaller than trees but have many hard stems near the ground. They are sometimes called bushes.
- *Examples:* Hibiscus plant, Rose plant, Cocoyam plant.
- **Herbs:** These are small plants with soft, green stems. They are usually short and do not live very long.
- *Examples:* Tomato plant, Maize plant, Spinach (Vegetable) plant, Beans plant.

Uses of Plants

Plants are very important to us. We depend on them for many things.

- **Food:** We eat fruits (mango, orange), vegetables (okra, spinach), and grains (rice, maize) from plants.
- **Shelter:** We use wood from trees to build houses, make furniture (chairs, tables), and doors.
- **Clothing:** We get cotton from cotton plants to make our clothes, towels, and bedsheets.
- **Medicine:** Some leaves, roots, and barks are used to cure sickness (e.g., Neem leaves, ginger). Many modern medicines come from plants.
- **Fresh Air (Oxygen):** Plants give us clean air to breathe by taking in carbon dioxide and giving out oxygen.
- **Shade and Beauty:** Trees provide shade on sunny days. Flowers make our environment beautiful.

EVALUATION

1. Which type of plant is very big and tall?
2. Give one example of a tree.
3. Is a tomato plant a tree or a herb?
4. Mention three things we get from plants (e.g., food, wood).
5. What do we use to make furniture like chairs? (Wood from ____)
6. Why are plants good for the air? (They give us oxygen)

ASSIGNMENT

1. Look around your house or school. Name one tree you see.
 2. Draw a simple chair made of wood.
 3. List two foods you ate today that came from plants.
-



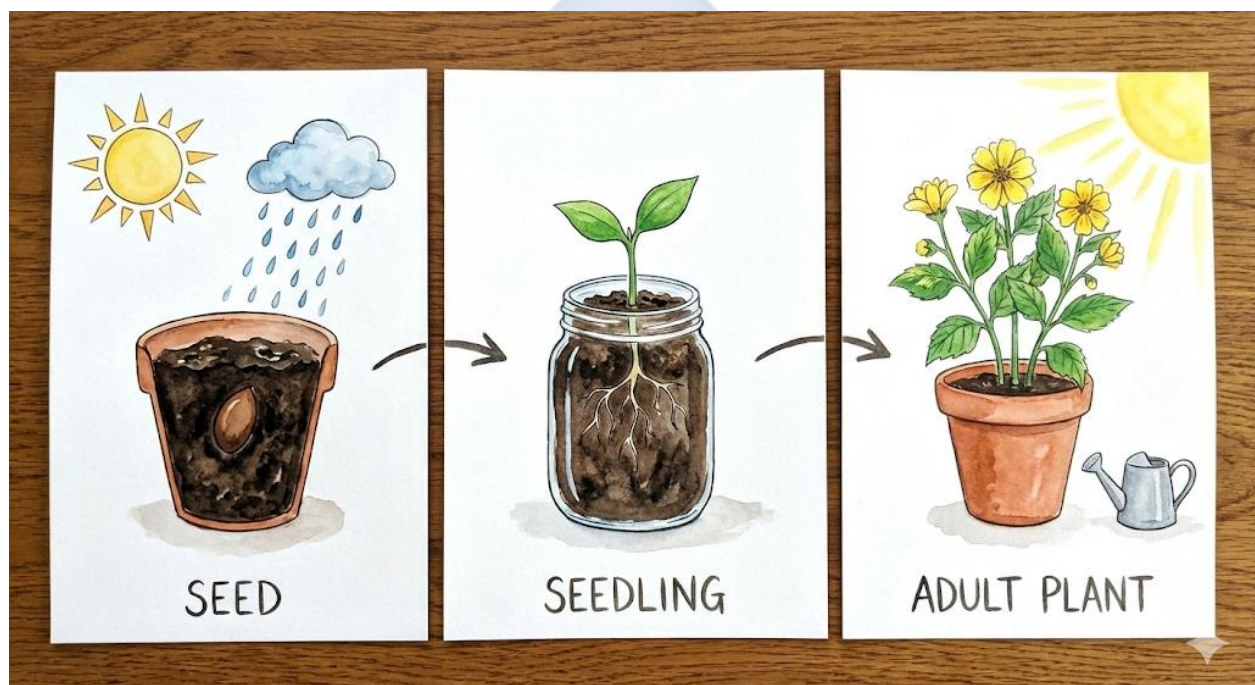
WEEK 8: PLANTS (PART 3): HOW PLANTS GROW.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- List four things plants need to grow.
- Describe the stages of plant growth from seed to adult.
- State two ways to care for plants.

CONTENT



What Plants Need to Grow

Just like children need food and water to grow, plants need things too. A plant will die without these four important things:

- **Soil:** To hold the roots and provide minerals (plant food).
- **Water:** To drink and to help carry food within the plant.
- **Air:** To breathe (they take in carbon dioxide and give out oxygen).
- **Sunlight:** To help them make their own food and keep warm. This is their energy.

Stages of Plant Growth

Plants usually start life as a seed. The growth stages are: SEED --> SEEDLING --> ADULT PLANT (with flowers/fruits)

- **Seed:** We put the seed in the soil. The seed is like a baby plant sleeping in a coat.
- **Germination:** The seed cracks open when it gets water and warmth. A tiny root comes out first, then a shoot.
- **Seedling:** The tiny baby plant comes out of the soil. It has small leaves. We must care for it gently.
- **Adult Plant:** The seedling grows bigger, grows more leaves, then flowers, and finally fruits. The fruits have new seeds inside.

Caring for Plants We must help plants grow well. This is called **cultivation**.

- **Watering:** Pour water on them when the soil is dry. But do not give too much water.
- **Weeding:** Remove unwanted grass and weeds near the plant because they steal water and food.
- **Protection:** Do not let goats, chickens, or other animals eat the plants. Use a fence if needed.
- **Support:** Some plants (like tomatoes) need sticks to hold them up.

EVALUATION

1. Mention four things a plant needs to grow.
2. What do we call a baby plant that just comes out of the soil?
3. Where do we put the seed to make it grow?
4. Why do plants need sunlight? (To make food)
5. What is weeding? (Removing unwanted grass)
6. What happens if you don't water a plant? (It wilts and may die)

ASSIGNMENT

1. Draw a seed and a seedling (baby plant) next to it.
2. With your parent's help, plant a bean seed in a small tin with soil. Water it lightly.

3. Write down one way you will care for your bean plant (e.g., water it daily).



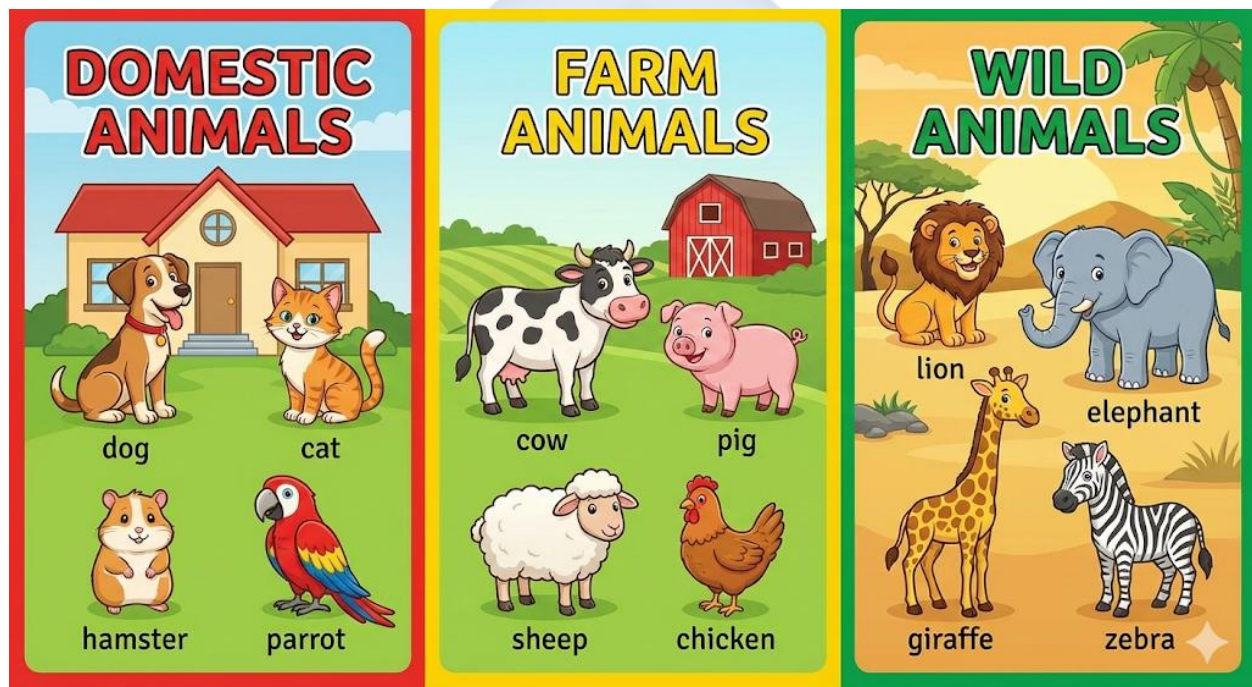
WEEK 9: ANIMALS (PART 1): TYPES OF ANIMALS.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Define animals as living things that move.
- Identify and list domestic (pet) animals.
- Identify and list farm animals.
- Identify and list wild animals.

CONTENT



Understanding Animals

Animals are living things. Unlike plants, animals can move from one place to another on their own to look for food, water, and shelter.

Types of Animals

We group animals based on where they live and how they live with humans.

- **Domestic Animals (Pets):** These animals live with us inside our homes or compounds. They are tame and our friends.
- *Examples:* Dog, Cat, Parrot, Rabbit, Canary bird.
- We care for them, and they provide companionship and sometimes security.
- **Farm Animals:** These animals are kept by farmers in a farm or pen. They are raised for food, work, or products.
- *Examples:* Cow, Goat, Sheep, Chicken, Pig, Duck, Turkey.
- They provide us with meat, milk, eggs, and wool.
- **Wild Animals:** These animals live freely in the bush, forest, jungle, or sea. They are not tame and can be dangerous. We mostly see them at the zoo, in books, or on TV.
- *Examples:* Lion, Elephant, Monkey, Snake, Crocodile, Tiger, Giraffe.

EVALUATION

1. Can animals move from place to place on their own? (Yes/No)
2. Where do domestic animals (pets) live? (With humans/At home)
3. Mention two examples of pets.
4. Mention two examples of farm animals.
5. Where does a lion live? (Forest/Jungle)
6. Is a snake a domestic animal? (Yes/No)

ASSIGNMENT

1. Draw your favourite animal and colour it.
 2. Name one animal that lives in the forest (wild animal).
 3. Name one farm animal that gives us eggs.
-

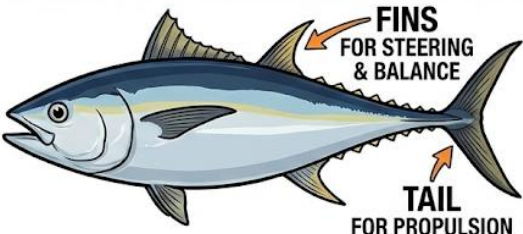
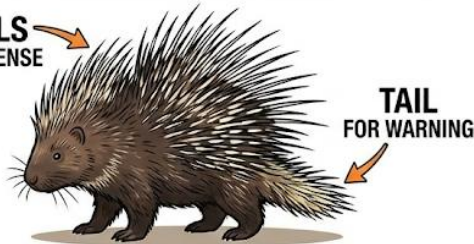
WEEK 10: ANIMALS (PART 2): BODY PARTS AND USES OF ANIMALS.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Identify common body parts of animals.
- Mention special body parts for birds and fish.
- List three uses of animals to humans.
- State two ways to care for animals.

CONTENT

ANIMAL BODY PARTS & THEIR USES	
 WINGS FOR FLIGHT & HUNTING TALONS FOR CATCHING PREY USE: PROTECTION & FOOD	 FINS FOR STEERING & BALANCE TAIL FOR PROPULSION USE: FOOD SOURCE
 LEGS FOR MOVEMENT & STRENGTH HOOVES FOR SUPPORT USE: WORK & TRANSPORT	 QUILLS FOR DEFENSE TAIL FOR WARNING USE: PROTECTION

Body Parts of Animals

Most animals have similar body parts:

- **Head:** (with eyes, ears, nose, mouth)
- **Body (Trunk)**
- **Legs:** (for walking and running) - some have 2, 4, or more legs.
- **Tail:** (for balance or swatting flies)

Special Body Parts

Different animals have special parts to help them move and survive in their environment.

- **Birds:** Have **wings** to fly and a **beak** (no teeth) to eat. They have feathers.
- **Fish:** Have **fins** and a **tail** to swim in water. They have gills to breathe underwater.
- **Snakes:** Have no legs; they slither on the ground using their muscles.
- **Insects (e.g., butterflies):** Have 6 legs and antennae.

Uses of Animals

Animals help us in many ways:

- **Food:** We get meat (cow, goat), milk (cow, goat), and eggs (chicken, duck) from animals.
- **Work:** Donkeys, horses, and camels carry heavy loads. Oxen help farmers plow the ground.
- **Protection:** Dogs guard our houses from thieves. Some dogs help police catch criminals.
- **Clothing and Materials:** We use leather (cow skin) for shoes, bags, and belts. We use wool (sheep hair) for sweaters and blankets.
- **Companionship:** Pets like cats and dogs are our friends and make us happy.

Caring for Animals

If we keep animals, we must treat them kindly.

- Give them clean food and fresh water every day.
- Build a clean, safe place (shelter) for them to sleep and rest.
- Take them to the animal doctor (veterinarian) when they are sick.
- Do not beat or hurt animals. They can feel pain too.

EVALUATION

1. What do birds use to fly? (Wings)
2. What do fish use to swim? (Fins and Tail)
3. Mention one animal we eat as meat.

4. What animal can guard our house?
5. What do we get from a sheep? (Wool/Meat)
6. How should we treat animals when they are sick? (Take them to the animal doctor)

ASSIGNMENT

1. List three body parts of a dog (e.g., head, legs, tail).
 2. Draw a simple fish and show the fins.
 3. Write one rule for caring for a pet dog (e.g., give it water daily).
-



WEEK 11: REVISION.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Recall the states of water and the water cycle.
- Identify primary and secondary colours.
- Describe parts and uses of plants.
- Differentiate between domestic and wild animals.

CONTENT

Review of Water

- **States:** Liquid (water), Solid (ice), Gas (steam).
- **Cycle:** Rain → River → Sun (evaporation) → Cloud → Rain again.

Review of Colours

- **Primary:** Red, Blue, Yellow (cannot be mixed from others).
- **Secondary:** Green (Blue+Yellow), Orange (Red+Yellow), Purple (Red+Blue).

Review of Plants

- **Parts:** Roots, Stem, Leaves, Flower, Fruit.
- **Uses:** Food, Wood, Medicine, Oxygen, Beauty.

Review of Animals

- **Pets (Domestic):** Live with us (Cat, Dog).
- **Farm Animals:** Kept for products (Cow, Chicken).
- **Wild Animals:** Live in forest (Lion, Elephant).
- **Uses:** Food, Work, Protection, Materials.

EVALUATION

- Teacher asks random questions from all topics taught during the term.

- **Example:** What color is the sun? (Yellow)
- What does a fish use to swim? (Fins)
- Is ice hot or cold? (Cold)
- Name one primary colour. (Red)

ASSIGNMENT

1. Study your notes from Week 1 to Week 10.
 2. Practice drawing a simple plant and an animal.
 3. Prepare for the examination (get your pencils and eraser ready).
-



PRIMARY 2 BASIC SCIENCE SCHEME OF WORK

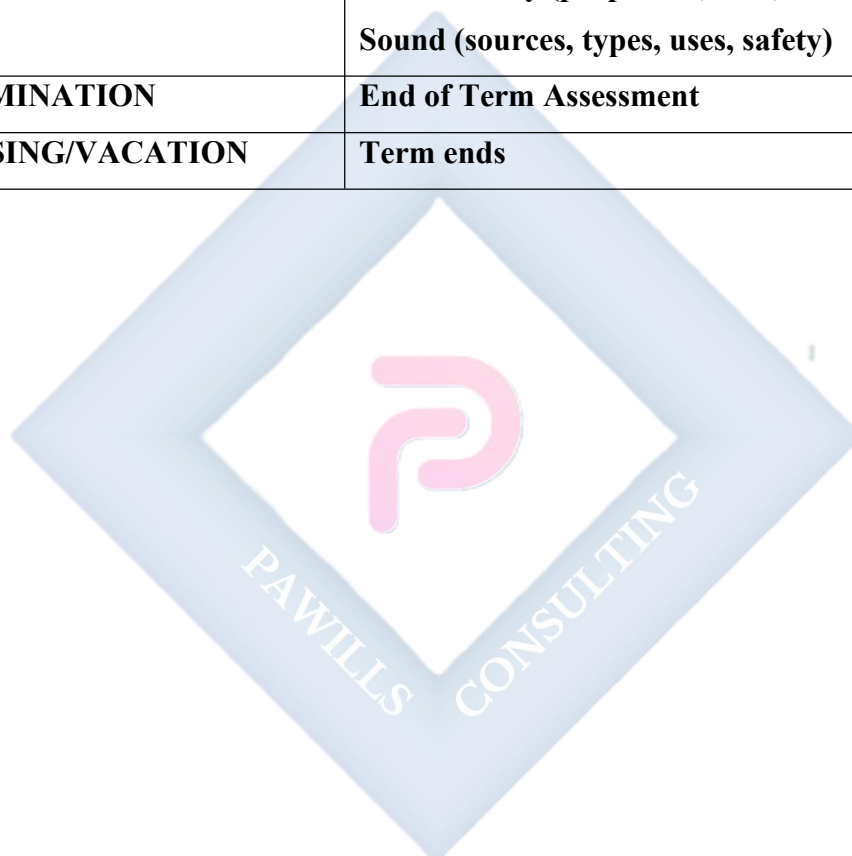
THIRD TERM

Week	Topic	Breakdown (Subtopics)
1	Clay (Part 1): What is Clay?	• Understanding clay: special type of soil that is sticky when wet • Where clay is found: near rivers, in some farms, clay pits • Properties of clay: - Soft and moldable when wet - Can be shaped into different forms - Becomes hard when dry - Smooth texture • Difference between clay and other soils • Activity: Touching and feeling wet clay, observing dry clay
2	Clay (Part 2): Uses of Clay	• What we make with clay: - Pots for cooking and storing water - Plates and bowls for serving food - Bricks for building houses - Tiles for floors and walls - Decorative items: vases, sculptures - Children's toys • People who work with clay: potters • Traditional pottery in Nigeria: different ethnic groups' pottery styles • Activity: Visiting a potter (if possible) or watching video of pottery making
3	Clay (Part 3): Working with Clay	• How to prepare clay for molding: adding water, kneading • Simple clay modeling techniques:

		- Rolling: making ropes and balls - Pinching: making small pots - Flattening: making flat shapes - Joining pieces together • Drying clay objects: leaving in sun or shade • Safety with clay: washing hands before and after, not eating clay • Activity: Simple clay modeling - making balls, ropes, simple pots, animals
4	Clay (Part 4): Clay Art and Projects	• Decorating clay objects: making patterns, adding textures • Painting dried clay objects • Creating clay models: fruits, animals, simple shapes • Appreciating traditional Nigerian pottery • Caring for clay objects: handling gently, keeping dry • Activity: Creating clay art project (individual or group), decorating clay objects
5	MID-TERM EXAMINATION	Assessment of Weeks 1-4
6	MID-TERM BREAK	Holiday/Rest
7	Forms of Energy: Sound (Part 1)	• Understanding energy: what makes things work, move, or change • Introduction to sound: what we hear with our ears • How sound is made: by vibrations (things shaking/moving quickly) • Sources of sound: - Our voice: vocal cords vibrate - Musical instruments: drums, bells, whistles - Animals: dogs bark, birds sing, cats meow

		- Things around us: doors closing, car engines, rain • Activity: Making sounds with our voices, identifying sounds
8	Forms of Energy: Sound (Part 2)	• Types of sounds: - Loud sounds: drum, thunder, car horn, shouting - Soft/quiet sounds: whisper, breeze, tiptoe - Pleasant sounds: music, singing, birds - Unpleasant sounds: noise, screaming, grinding • High and low sounds (pitch): bird singing (high), drum beating (low) • Fast and slow sounds • Activity: Creating loud and soft sounds, identifying pleasant and unpleasant sounds
9	Forms of Energy: Sound (Part 3)	• How sound travels: through air, can travel through water and solid objects • Sound travels from source to our ears • Sounds can be near (close) or far (distant) • Echo: hearing sound again (like in empty room or mountain) • Uses of sound: - Communication: talking, telephone, school bell - Warning: alarm, siren, car horn - Entertainment: music, songs, stories • Activity: Sound experiments - sounds through different materials
10	Forms of Energy: Sound (Part 4) - Musical Instruments and Sound Safety	• Making musical instruments: - Shakers: containers with beans/stones - Drums: containers with stretched material - Bells: metal objects that ring - Whistles: blowing instruments • Traditional Nigerian instruments: drums, gong,

		shekere, flute • Sound safety: - Very loud sounds can hurt our ears - Don't shout in people's ears - Cover ears when near very loud sounds • Activity: Making simple musical instruments from recycled materials, class orchestra
11	REVISION	Review: Clay (properties, uses, modeling), Sound (sources, types, uses, safety)
12	EXAMINATION	End of Term Assessment
13	CLOSING/VACATION	Term ends



PRIMARY 2 BASIC SCIENCE LESSON NOTES

THIRD TERM

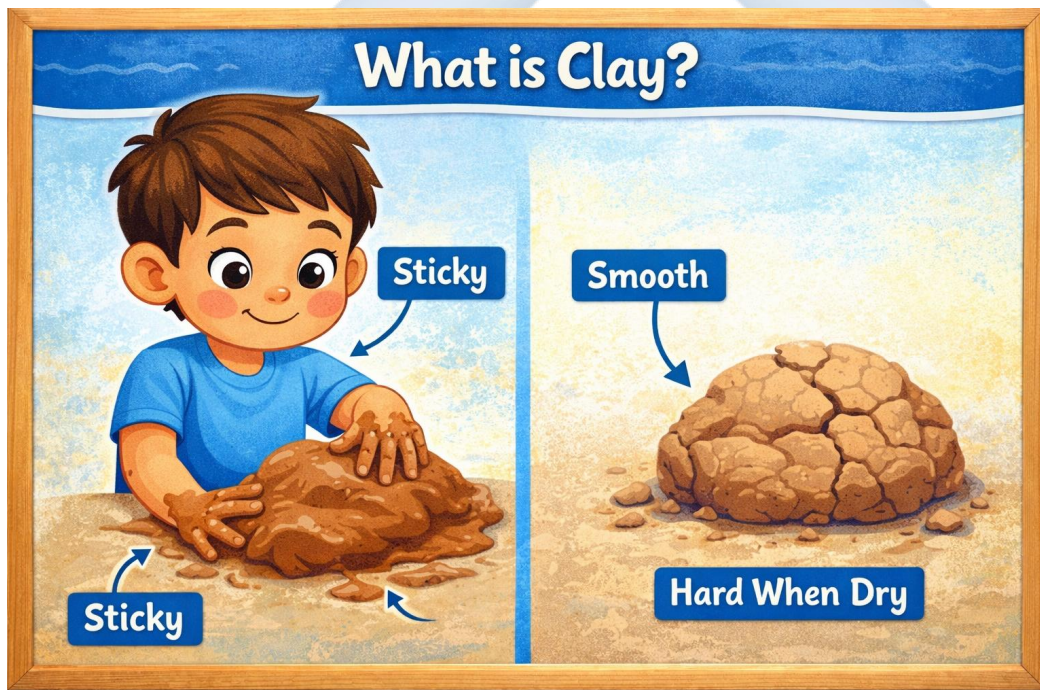
WEEK 1: CLAY (PART 1): WHAT IS CLAY?.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Define clay as a special type of soil.
- State where clay can be found.
- Mention three properties of clay.
- Differentiate between clay and sandy soil.

CONTENT



Meaning of Clay

Clay is a special type of soil with very tiny, fine particles. It is different from the sand we see on the playground.

- When clay is wet, it is very sticky and can be shaped.

- When it is dry, it becomes very hard like a stone.

Where Clay is Found

We can find clay in specific places such as:

- Near river banks (beside the river) where the water has deposited fine soil.
- In clay pits (holes dug deep in the ground).
- On some farms or swampy areas where the soil is always wet.
- In some backyards, especially after heavy rain.

Properties of Clay

How do we know a soil is clay?

- **Texture (Feel):** It feels smooth and slippery when wet. When dry, it feels powdery and hard.
- **Moldable:** You can shape it into a ball, a snake, or a pot easily when wet. It holds its shape.
- **Water Holding:** It holds water very well and becomes sticky. It does not let water pass through quickly.
- **Color:** It can be grey, brown, reddish, or even white.
- **Hardness:** When you leave it in the sun, it dries and becomes as hard as a brick. It may crack.

Difference Between Clay and Sand

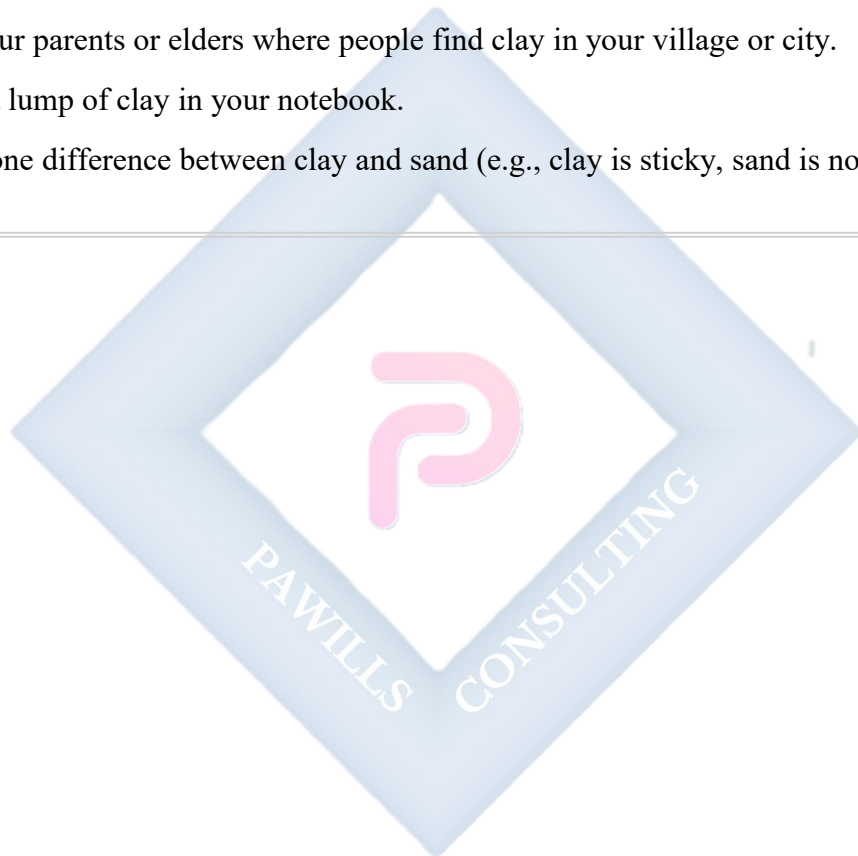
CLAY	SAND
Sticky when wet	Gritty and rough
Holds water	Water drains fast
Can be molded	Cannot hold shape
Smooth texture	Grainy texture

EVALUATION

1. What is clay? (A type of sticky soil)
2. Where can we find clay? (Near rivers, in pits)
3. How does clay feel when it is wet? (Sticky/Smooth)
4. What happens to clay when it dries in the sun? (It becomes hard)
5. Which soil is sticky: Sand or Clay?

ASSIGNMENT

1. Ask your parents or elders where people find clay in your village or city.
 2. Draw a lump of clay in your notebook.
 3. Write one difference between clay and sand (e.g., clay is sticky, sand is not).
-



WEEK 2: CLAY (PART 2): USES OF CLAY.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- List four items made from clay.
- Identify a potter as someone who works with clay.
- Mention the use of bricks and tiles.
- Describe how clay is used in Nigerian tradition.

CONTENT



What We Make with Clay

Clay is very useful because it can be shaped into many things and then hardened.

- **Household Items:** We use clay to make pots for cooking (e.g., "Ikoko" for stew), pots for cooling and storing water, plates, bowls, and cups.
- **Building Materials:** Clay is used to make strong bricks for building houses and tiles for floors and roofs.
- **Art and Toys:** We make beautiful sculptures, flower vases, and toys for children (like clay cars or animals).

- **Other Uses:** Clay is used to make local money boxes (piggy banks), ceramic sinks, and toilet bowls.

People Who Work with Clay

A person whose job is to make things with clay is called a **Potter**. They use their hands, a turning wheel, or simple tools to shape the clay into beautiful and useful objects.

Traditional Pottery in Nigeria In Nigeria, many cultures have used clay for centuries.

- We use big clay pots ("Omu" or "Ganga") to store water so it stays cool.
- We use clay pots to cook soup because it gives the food a special taste.
- Clay is used to make local stoves ("Matafiri").
- In some cultures, clay pots are used for rituals and ceremonies.

EVALUATION

1. Who is a potter? (A person who makes things from clay)
2. Mention two things we use clay to make in the kitchen.
3. What is used to build strong walls of houses? (Bricks)
4. Why do people store water in clay pots? (To keep it cool)
5. Can we use clay to make toys? (Yes/No)

ASSIGNMENT

1. Draw a traditional clay pot used for fetching water.
 2. Draw a simple house made with bricks.
 3. List three things in your house that can be made of clay (e.g., flower vase, brick, tile).
-

WEEK 3: CLAY (PART 3): WORKING WITH CLAY.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Explain how to prepare clay for molding.
- Describe "kneading" of clay.
- Demonstrate simple molding techniques like rolling and pinching.
- State safety rules when working with clay.

CONTENT



Preparing Clay

Before we can make things, we must prepare the clay to be soft and smooth.

- **Dig:** Get the clay from the ground.
- **Clean:** Remove stones, sticks, leaves, and other dirt.
- **Mix:** Add a little water to the dry clay and mix until it is like soft dough. Not too wet, not too dry.

- **Knead:** Press, squeeze, and fold the clay with your hands (like making bread dough) for several minutes. This removes air bubbles and makes the clay smooth and easy to shape.

Simple Modeling Techniques

- **Rolling:** Putting clay between your palms or on a flat surface and moving your hands back and forth to make a long shape like a snake or a rope.
- **Pinching:** Using your thumb and fingers to press a hole in a ball of clay and then pinching the sides to make a small pot or bowl.
- **Flattening:** Pressing the clay with your palm or a rolling pin to make a flat shape like a pancake, tile, or leaf.

Drying

After making an object, we leave it in a shady place to dry slowly for a day or two. If we dry it too fast in the sun, it may crack. When completely dry, it becomes hard and brittle.

Safety with Clay

- **Do not eat clay:** It is not food and can make you sick.
- **Do not throw clay:** at your friends. It can hurt them or make a mess.
- **Always wash your hands:** with soap and water after playing with clay.
- **Clean your work area:** after finishing.

EVALUATION

1. What do we add to dry clay to make it soft? (Water)
2. What is kneading? (Pressing and squeezing clay to make it smooth)
3. Describe how to make a clay "snake." (Roll the clay between your hands)
4. Where should we dry our clay objects? (In a shady place)
5. Is it safe to eat clay? (No)

ASSIGNMENT

1. List the steps to make a clay ball (e.g., take a piece of clay, roll it in your hands).
2. Why must we wash our hands after using clay? (To remove germs and dirt)

3. Draw a picture of a clay snake.



WEEK 4: CLAY (PART 4): CLAY ART AND PROJECTS.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Describe how to decorate clay objects.
- List examples of clay art models (fruits, animals).
- Explain how to care for clay objects.
- Appreciate the beauty of Nigerian pottery.

CONTENT



Decorating Clay Objects

To make clay objects beautiful, we can decorate them before they dry completely.

- **Make Patterns:** Use a stick, toothpick, fork, or stone to draw lines, dots, and shapes on the wet clay.
- **Add Texture:** Press rough objects like leaves, lace, or nets onto the clay to leave an imprint.
- **Attach Pieces:** Add small pieces of clay to make handles, eyes, or decorations.
- **Paint:** When the clay is completely dry, we can use water-based paints or natural dyes to colour it.

Creating Clay Models

We can use clay to copy things we see around us. This is called modeling.

- **Fruits:** Oranges, bananas, apples, mangoes.
- **Animals:** Snakes, lizards, birds, fish, elephants.
- **Everyday Objects:** Cups, plates, cars, houses.
- **Shapes:** Stars, hearts, circles.

Caring for Clay Objects

Clay objects are fragile (they break easily) if they are not fired (baked in a special oven called a kiln).

- Handle them gently with two hands.
- Do not drop them on the floor.
- Keep them in a dry place so they don't get wet and turn back to mud.
- Dust them gently with a soft cloth.

Appreciating Nigerian Pottery

Nigeria has a rich tradition of pottery. Different ethnic groups make unique and beautiful pots, often with special patterns and uses. We should be proud of this art.

EVALUATION

1. Mention one way to make a clay pot beautiful. (Draw patterns/Paint)
2. Name two animals you can model with clay.
3. What happens if you drop a dry clay pot? (It may break)
4. How should you carry a clay object? (Carefully with two hands)
5. Do we paint the clay when it is wet or dry? (Dry)

ASSIGNMENT

1. Design a pattern (lines and dots) for a flower vase on paper.
2. List three fruits you would like to make with clay.
3. Draw a painted clay pot.



WEEK 5 & 6: MID-TERM EXAMINATION AND MID-TERM BREAK.

(Note: No lesson notes are required for these weeks as they are for assessment and holiday.)



WEEK 7: FORMS OF ENERGY: SOUND (PART 1).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Define sound as what we hear.
- Identify the ear as the sense organ for sound.
- Explain simply how sound is made (vibration).
- List four sources of sound.

CONTENT



Understanding Sound

Energy is what makes things work or move. Sound is a form of energy. Sound is anything we can hear with our ears.

How Sound is Made

Sound is made by **Vibration**. Vibration means shaking or moving back and forth very quickly.

- When you beat a drum, the skin of the drum shakes (vibrates) and pushes the air around it, creating sound waves.
- When you talk, your vocal cords in your throat vibrate.

- When a phone rings, a small part inside it vibrates.

Sources of Sound

Where does sound come from? Everywhere!

- **Human Beings:** Talking, singing, shouting, clapping, coughing.
- **Musical Instruments:** Drums, bells, flutes, pianos, guitars.
- **Animals:** Dogs barking, birds chirping, cats meowing, lions roaring.
- **Objects and Machines:** Cars honking, doors slamming, rain falling, phones ringing, televisions.

EVALUATION

1. What is sound? (What we hear/Form of energy)
2. Which part of the body do we use to hear?
3. How is sound made? (By vibration/shaking)
4. Mention two animals that make sounds.
5. Mention two objects in the classroom that can make a sound (e.g., clock, chair moving).

ASSIGNMENT

1. Touch your throat gently and hum a song. Describe what you feel. (Vibration)
 2. List three sources of sound you can hear at home.
 3. Draw a bell ringing.
-

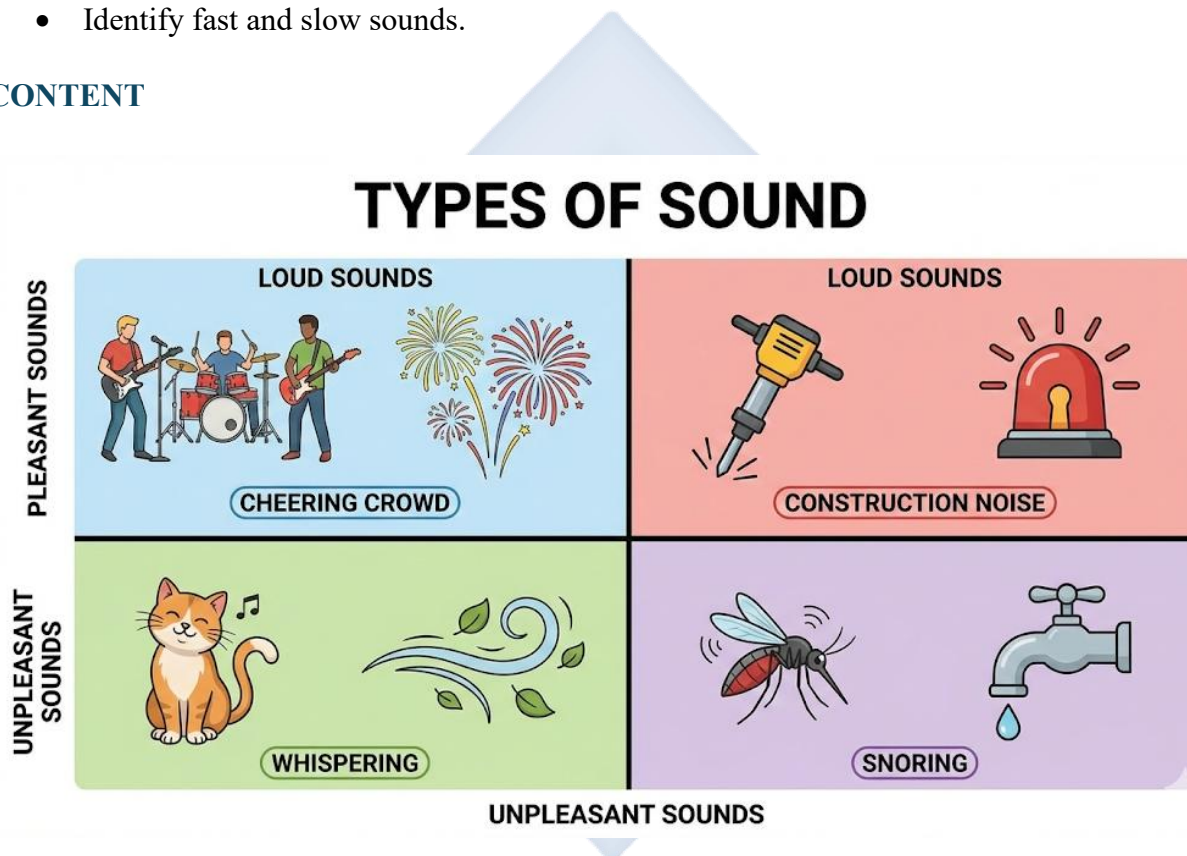
WEEK 8: FORMS OF ENERGY: SOUND (PART 2).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Differentiate between loud and soft sounds.
- Differentiate between pleasant and unpleasant sounds.
- Define pitch as high or low sound.
- Identify fast and slow sounds.

CONTENT



Types of Sounds: Loud and Soft

- **Loud Sounds:** These are strong, powerful noises that you can hear from far away.
- *Examples:* Thunder, drum beating hard, car horn, aeroplane, shouting.
- **Soft (Quiet) Sounds:** These are gentle, weak noises that you hear only when you are close or in a quiet place.

- *Examples:* Whispering, writing with a pencil, a drop of water, breathing, a mouse squeaking.

Pleasant vs. Unpleasant Sounds

- **Pleasant Sounds:** Sounds that are nice to hear and make us feel happy or relaxed.
- *Examples:* Music, singing, birds singing, gentle rain, a friend's laugh.
- **Unpleasant Sounds (Noise):** Sounds that disturb us, hurt our ears, or make us feel uncomfortable.
- *Examples:* Screaming, scratching a blackboard, generator noise, broken car exhaust, crying baby.

High and Low Sounds (Pitch) Pitch tells us if a sound is high or low.

- **High Sound (High Pitch):** Sharp and squeaky, like a small bird (chirp), a whistle, or a child's voice.
- **Low Sound (Low Pitch):** Deep and rumbling, like a big drum, a cow mooing, or a man's deep voice.

Fast and Slow Sounds

Sounds can also be fast or slow. This is about how quickly the vibrations happen.

- **Fast vibrations** make high sounds.
- **Slow vibrations** make low sounds.

EVALUATION

1. Give an example of a loud sound.
2. Give an example of a soft sound.
3. Is music a pleasant or unpleasant sound?
4. What do we call unpleasant sound? (Noise)
5. Does a whistle make a high sound or a low sound? (High)

ASSIGNMENT

1. List two pleasant sounds you like to hear.
 2. List two unpleasant sounds you dislike.
 3. Is thunder a soft sound? (Yes/No)
-



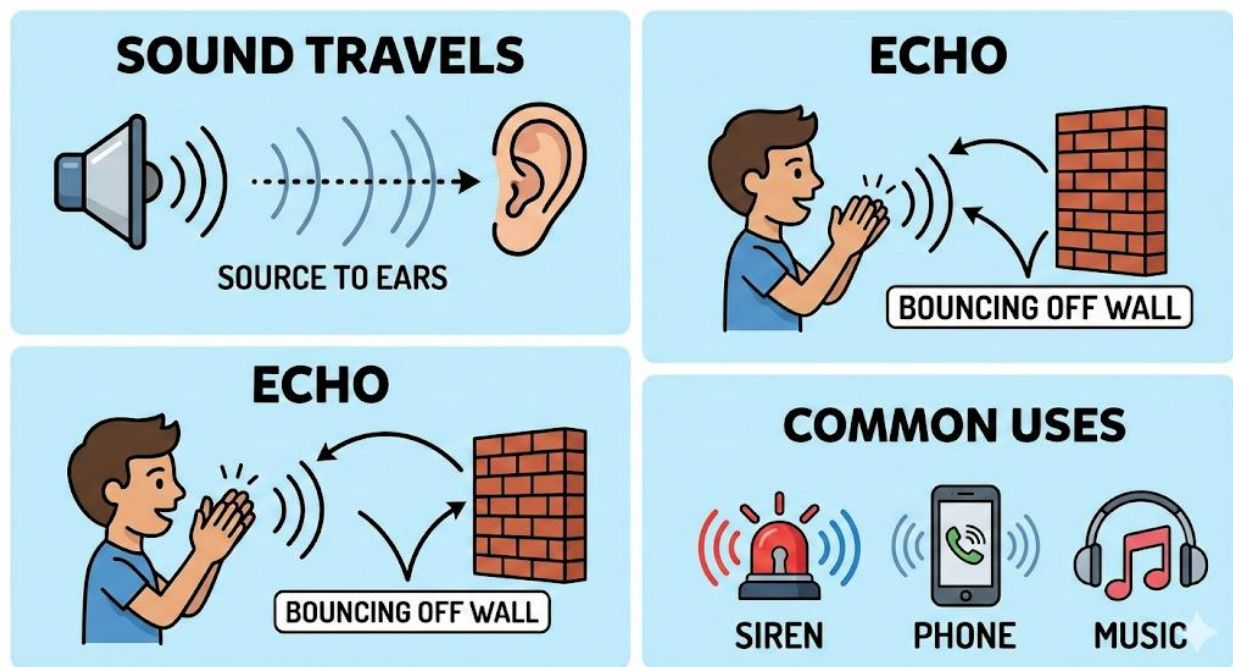
WEEK 9: FORMS OF ENERGY: SOUND (PART 3).

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Explain that sound travels through air.
- Define echo as a reflected sound.
- List three uses of sound.
- Distinguish between near and far sounds.

CONTENT



How Sound Travels

Sound moves from the source (where it starts) to our ears.

- Sound travels through the **air** in waves. That is how we hear a teacher speaking across the room.
- Sound can also travel through **water** (dolphins talk underwater) and **solid objects** (like hearing a knock on a door through the wall).

- Sound cannot travel through empty space (a vacuum). There is no air to carry the sound.

Near and Far Sounds

- If a sound is **loud and clear**, the object making the sound is likely **near**.
- If a sound is **faint (soft) and hard to hear**, the object is likely **far away**.

Echo

An **Echo** is when you hear a sound again after a short time. This happens when sound waves hit a hard, flat surface (like a wall or mountain) and bounce back to your ears.

- **Example:** Shouting in an empty hall or near a big building and hearing your voice come back.

Uses of Sound

- **Communication:** We talk to share information, ask questions, and tell stories. We use telephones and radios.
- **Warning and Safety:** Sirens (ambulance, police, fire truck), car horns, and alarm clocks tell us to pay attention or wake up.
- **Entertainment:** We listen to music, stories, and watch television for fun.
- **Notifications:** School bells tell us when to go to class or break. Doorbells tell us someone is at the door.
- **Medicine:** Doctors use stethoscopes to listen to sounds inside our body (heartbeat, lungs).

EVALUATION

1. What does sound travel through to reach our ears? (Air/Water/Solids)
2. What is an echo? (A sound that bounces back)
3. Mention one way we use sound for warning. (Sirens/Horns)
4. Mention one way we use sound for fun. (Music/Games)
5. If you hear a faint car horn, is the car near or far? (Far)

ASSIGNMENT

1. Draw a telephone (a tool for communication using sound).

2. Why do ambulances use sirens? (To warn people to give way)
 3. Stand outside your house, close your eyes, and listen for one minute. Write down two different sounds you hear.
-



WEEK 10: FORMS OF ENERGY: SOUND (PART 4) - MUSICAL INSTRUMENTS AND SOUND SAFETY.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Identify common musical instruments.
- Mention traditional Nigerian instruments.
- Explain how to make a simple instrument (shaker).
- State safety rules for protecting the ears.

CONTENT



Musical Instruments

These are objects made specially to produce pleasant, musical sounds when played.

- **Shakers (Maracas):** Containers filled with seeds, beads, or stones. You shake them to play.

- **Drums:** Made with a hollow body (wood, clay) and leather skin stretched over it. You beat them with hands or sticks.
- **Bells:** Metal objects that ring when hit or shaken.
- **Whistles:** You blow air into them to make sound.
- **Guitar/Piano:** Have strings or keys that vibrate to make music.

Traditional Nigerian Instruments

Nigeria has many beautiful traditional instruments.

- **Talking Drum (Gangan/Dundun):** Used in Yoruba land. Its pitch can be changed to mimic speech.
- **Gong (Ogene):** A metal instrument hit with a stick, used in Igbo culture.
- **Shekere:** A dried gourd covered with a net of beads. It is shaken or hit.
- **Flute (Oja):** A wooden wind instrument used for whistling music.
- **Udu:** A clay pot drum with a hole, played with the hands.

Making a Simple Shaker

You can make your own shaker at home:

1. Take a small, empty plastic bottle with a lid (e.g., water bottle).
2. Fill it with dry beans, rice, or small stones (about 1/4 full).
3. Close the lid tightly.
4. Decorate the outside with paper or paint.
5. Shake it to make music!

Sound Safety

We must protect our ears because they are delicate and can be damaged by very loud sounds.

- **Do not shout** directly into someone's ear. It can hurt their eardrum.
- **Avoid very loud noises** like standing near loudspeakers at parties, or listening to music at full volume with headphones.

- **Cover your ears** with your hands or use earplugs if a sound is too loud (like during fireworks).
- **See a doctor** if your ear hurts or you have trouble hearing.

EVALUATION

1. Mention two musical instruments.
2. What is inside a shaker to make it sound? (Seeds/Stones)
3. Name one traditional Nigerian instrument.
4. Why should we not shout in someone's ear?
5. What should you do if you hear a very loud noise? (Cover your ears)

ASSIGNMENT

1. Draw a drum.
2. List three materials you can use to make a simple shaker at home (e.g., bottle, beans, tape).
3. Write one rule to protect your ears from loud noise.

WEEK 11: REVISION.

OBJECTIVES

By the end of the lesson, pupils should be able to:

- Recall the properties and uses of clay.
- Recall the definition, sources, and types of sound.
- Answer questions covering the term's work confidently.

CONTENT

Review of Clay

- **What is it?** Sticky soil found near rivers, in pits.
- **Properties:** Smooth, moldable, hard when dry.
- **Uses:** Pots, bricks, tiles, toys.
- **Working with clay:** Knead, mold, dry, decorate.

Review of Sound

- **Definition:** A form of energy we hear, made by vibration.
- **Sources:** Humans, animals, instruments, machines.
- **Types:** Loud/Soft, Pleasant/Unpleasant (Noise), High/Low pitch.
- **Travel:** Through air, water, solids.
- **Uses:** Communication, warning, entertainment.
- **Safety:** Protect ears from loud noises.

EVALUATION

- Teacher asks mixed questions from Week 1 to Week 10.
- **Example:** What is a potter? (Person who works with clay)
- How is sound made? (By vibration)
- Is clay rough or smooth when wet? (Smooth and sticky)
- Name a warning sound. (Siren, alarm)

ASSIGNMENT

1. Study all your notes from Week 1 to Week 10.
2. Prepare your writing materials (pencils, eraser) for the examination.
3. Practice drawing a clay pot and a musical instrument (like a drum).

